

■ Homework Excuse Generator

Vibe Coding Assignment — Prompt Documentation

Subject: Programming Fundamentals | Tool Used: Claude (Anthropic) | Language: C++

1. Assignment Overview

This assignment required building a console-based C++ program that generates creative, random excuses for missing homework or assignments. The program must accept a student's name, store at least 10 excuse templates, randomly select one each run, replace the {name} placeholder with the actual input, and handle edge cases like empty input. Students must use an AI Vibe Coding tool and document all prompt interactions, including a mandatory bug-fixing session.

2. Prompt Interactions with Claude

Prompt 1 — Initial Program Generation

■ Student prompt sent to Claude:

Write a beginner-friendly C++ console program that generates random student excuses for missing homework. The program should ask for the student's name, store at least 10 excuse templates in a vector, and randomly select one each time it runs. Replace the placeholder {name} with the user's actual input. Seed the random number generator with time() so excuses change every run. Add comments so beginners can follow the logic.

■ What Claude generated:

- Complete C++ program with #include for <iostream>, <vector>, <cstdlib>, <ctime>.
- A vector<string> holding 10 excuse templates with {name} placeholders.
- srand(time(nullptr)) to seed the RNG for different results every run.
- rand() % excuseTemplates.size() to pick a random template index.
- String replacement using std::string::find() and replace().
- Basic cin input to read the student's name.

Prompt 2 — Improving Excuse Creativity

■ Student prompt sent to Claude:

The excuse templates are too plain. Rewrite all 10 to be funnier and more creative — include tech failures, pet disasters, and absurd coincidences that students can relate to. Keep the {name} placeholder in every template.

■ Sample improved templates Claude provided:

"{name} couldn't complete the assignment because the laptop decided to install updates for six hours."

"{name} completed the assignment but the submission portal said 'file too large' – it was a Word doc."

"{name} tried to email the assignment but autocorrect changed the teacher's address to a pizza place."

```
"{name} was going to submit last night but got stuck in a two-hour loop of 'just one more YouTube video'."
```

Prompt 3 — Bug Identification (Empty Input & cin Issue)

■ Bug Observed:

Two bugs were found after first running the program. First, pressing Enter without typing a name caused the program to print an excuse with a blank space where the name should appear, e.g. ' **couldn't complete the assignment...**'. Second, using `cin >> studentName` would only capture the first word of a multi-word name like 'Priya Sharma', and also broke any subsequent `getline` calls by leaving a stray newline in the input buffer.

Buggy code snippet:

```
string studentName;
cout << "Enter your name: ";
cin >> studentName; // BUG 1: breaks on multi-word names
// BUG 2: no empty-string check
```

■ Bug-fix prompt sent to Claude:

My program has two bugs: (1) When the user presses Enter without typing anything, the excuse prints with a blank name — it should display an error and stop. (2) Using `cin >> studentName` only reads one word, so 'Priya Sharma' becomes just 'Priya'. Please fix both bugs and explain why `getline` is the better choice here.

■ Claude's corrected code:

```
getline(cin, studentName); // reads full name including spaces

if (studentName.empty()) {
    cout << "[Error] Name cannot be empty!\n";
    return 1; // exit gracefully
}
```

- **Why `getline`?** Unlike `cin >>`, `getline` reads the entire line including spaces, so full names like 'Priya Sharma' are captured correctly.
- **Empty check:** Testing `studentName.empty()` immediately after input prevents a blank name from reaching the template replacement logic.

Prompt 4 — Preventing Consecutive Repeats

■ Student prompt sent to Claude:

When I ask for another excuse, the same one sometimes appears twice in a row. Add logic so the program never picks the same excuse consecutively.

■ Claude's solution:

```
int getRandomExcuse(int total, int lastIndex) {
    if (total == 1) return 0;
    int idx;
    do { idx = rand() % total; } while (idx == lastIndex);
    return idx;
}
```

A do-while loop regenerates the index until it differs from the previous one, guaranteeing variety without any complex data structures.

Prompt 5 — Refactor & Final Polish

■ Student prompt sent to Claude:

Refactor the code: move the placeholder replacement into its own function called `replacePlaceholder()`. Add a loop so the user can request multiple excuses without restarting. Also add a welcome banner at the start and a fun closing message.

■ Final structure Claude produced:

- `replacePlaceholder(string templ, const string& name)` — replaces all {name} occurrences with a while loop.
- `getRandomExcuse(int total, int lastIndex)` — non-repeating random selection.
- `main()` — handles input validation, excuse display, and the repeat loop.
- Welcome banner and closing message for a polished terminal experience.

3. Prompt Quality Summary

#	Prompt Purpose	Clarity	Iteration	Outcome
1	Initial generation	High	1st attempt	Working base program
2	Improve excuse creativity	High	Refinement	Funnier, relatable excuses
3	Bug fix (empty + cin)	High	Bug report	Robust input validation
4	No repeat excuses	Medium	Refinement	Non-consecutive RNG
5	Refactor & polish	High	Final pass	Clean, modular code

4. Final Program Overview

The final `HomeworkExcuseGenerator.cpp` includes all improvements from the five prompts above:

- 12 creative, student-relatable excuse templates in a `std::vector<string>`.
- Random selection seeded with `srand(time(nullptr))` for unique results every run.
- `replacePlaceholder()` correctly replaces all {name} occurrences.
- Empty-input guard prints an error and exits cleanly with `return 1`.
- `getline` used instead of `cin >>` to support multi-word names.
- Repeat loop lets the user request additional excuses without restarting.
- Non-repeating RNG via `getRandomExcuse()` with do-while logic.
- Fully commented for beginner readability.